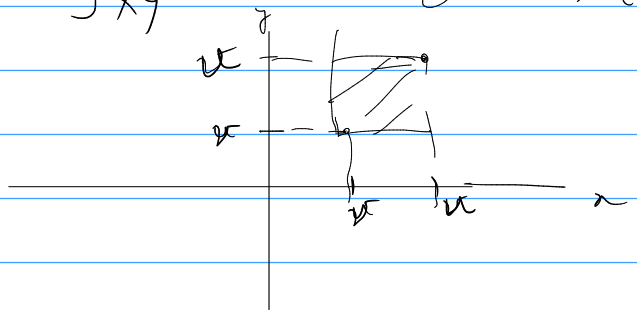
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$X, Y \sim f_{XY}$ $\quad U = \max(X,Y)$ $\quad V = \min(X,Y)$



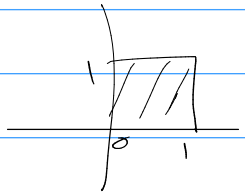
$F_{UV}(u,v) = Pr(\min(X,Y) \le v, \max(X,Y) \le u)$

$= F_{XY}(u,v) + F_{XY}(v,u) - F_{XY}(v,v)$

$f_{UV}(u,v) = \frac{\partial^2 F_{UV}}{\partial u \partial v} = f_{XY}(u,v) + f_{XY}(v,u)$

(3) $U = X + Y$ $\quad X, Y \sim$ iid Uniform $[0,1]$

$V = Y$



$f_U(u) = \int_{v=0}^{1} f_{XY}(u-v, v)\, dv$

$= \int_0^1 f_X(u-v) f_X(v)\, dv$

$u \in [0,2]$

dist of sum = conv of dist

indiv.

iid RVs

(4) $U = X/Y$ $\quad X, Y \sim$ iid $\frac{1}{\sqrt{\pi}} e^{-x^2/2}\, dx$

$V = Y$

$f_U(u) = \int_{v=-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{(u^2v^2+v^2)}{2}} \cdot |v|\, dv = \frac{1}{\pi(1+u^2)}$, Cauchy

$u \in \mathbb{R}$ $\quad u \in \mathbb{R}$

Multivariate distributions

$(X_1, X_2, \ldots, X_n)$: jointly distributed

All discrete

$p_{X_1 \ldots X_n}(x_1, \ldots, x_n)$

$S \subseteq [n] \triangleq \{1, 2, \ldots, n\}$

$p_{X_S}(x_S) = \sum_{x_{S^c}} p_{X_1 \ldots X_n}(x_1, \ldots, x_n)$

$(X_i : i \in S)$

Jointly continuous

$f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n)$

$f_{X_S}(x_S) = \int \cdots \int_{x_{S^c}} f_{X_1 \ldots X_n}(x_1, \ldots, x_n)\, dx_1 \ldots dx_n$

$$p_{X_S|X_T=x_T}(x_S) = \frac{p_{X_{S\cup T}}(x_{S\cup T})}{p_{X_T}(x_T)}$$

$S, T \subseteq [n]$

$S \cap T = \emptyset$

iid $X_1, X_2, \ldots, X_n \sim$ iid "some distribution"

iid p_X

iid f_X

iid X

Ex: (1) $X_1, X_2, \ldots, X_n \sim$ iid Bernoulli(p)

$Y = X_1 + \cdots + X_n \sim$ Binomial(n, p)

(2) $X_1, X_2, \ldots, X_k \sim$ iid Geom(p)

$X_i \in \{1, 2, 3, \ldots\}$ p $(1-p)p$ $(1-p)^2 p$ ...

$S_2 = X_1 + X_2 \in \{2, 3, \ldots\}$

$$\Pr(S_2 = m) = m p^2 (1-p)^{m-2}$$

$S_k = X_1 + X_2 + \cdots + X_k \in \{k, k+1, \ldots\}$

$$\Pr(S_k = m) = \binom{m-1}{k-1} p^k (1-p)^{m-k}$$

(3) N : random 4-digit number

X_i : i-th digit

$X_1 \sim$ Uniform$\{1, \ldots, 9\}$

$X_i \sim$ Uniform$\{0, 1, \ldots, 9\}$ $i = 2, 3, 4$

X_1, X_2, X_3, X_4 : indep.

$$p_{X_1 X_2 X_3 X_4} = p_{X_1} p_{X_2} p_{X_3} p_{X_4}$$

$Y_1 = I(N \text{ is even})$ $\quad Y_2 = I(N \text{ is a multiple of } 4)$

indicator

$Y_3 =$ " " 6

A: event

$I(A) = \begin{cases} 1, & \text{if } A \text{ occurs} \\ 0, & \text{else} \end{cases}$ $\quad Y_4 = I(N \text{ is prime})$

④ Trinomial $\quad X_1, X_2, \ldots, X_n \sim iid \begin{Bmatrix} p_0 & p_1 & p_2 \\ 0, & 1, & 2 \end{Bmatrix}$

$p_0 + p_1 + p_2 = 1$

$Y_0 = \#\,0\text{'s}, \quad Y_1 = \#\,1\text{'s}, \quad Y_2 = \#\,2\text{'s}$

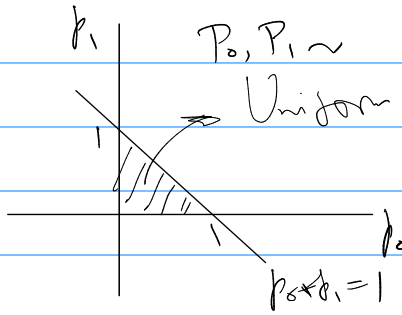
$$p_{Y_0, Y_1, Y_2}(a, b, c) = \begin{cases} \frac{n!}{a! \cdot b! \cdot c!} p_0^a p_1^b p_2^c, & a+b+c = n \\ 0, & a+b+c \neq n \end{cases}$$

$a, b, c > 0$ integers

$Y_0 \sim Binom(n, p_0)$

$Y_1 \mid Y_0 = y_0 \sim Binom\left(n - y_0, \frac{p_1}{p_1 + p_2}\right)$

⑤



$P_2 = 1 - P_0 - P_1$

$0 \leq P_0 + P_1 \leq 1$

(P_0, P_1, P_2): PMF on $\{0, 1, 2\}$

Given $P_0 = p_0, P_1 = p_1, P_2 = p_2$,

$X_1, X_2, \ldots, X_n \sim iid \begin{Bmatrix} p_0 & p_1 & p_2 \\ 0, & 1, & 2 \end{Bmatrix}$

$f_{P_0 \mid Y_0 = y_0}(p_0) = ?$